with the Southern Rhodesian National Archives and with the librarians of all three territories to supply micro-copies of scientific papers to the experimental stations—many of which are far from any substantial library facilities.

The Agricultural Research Council of Central Africa takes the assets, liabilities and agreements of the former Agricultural Research Council of Rhodesia and Nyasaland and will have substantial continuation of membership. Sir Ronald Prain has thus handed over to his successor an active organization which has a unique opportunity to play a constructive part in the development of all three territories. The Agreement by the three Governments, that scientific research lies outside politics and that biological problems of concern to the well-being of all the peoples in the area are best attacked jointly, offers a unique scientific opportunity and a sound vehicle by which well-wishers in more advanced countries may help in the development of Africa.

H. C. PEREIRA

PALÆOMAGNETIC SUPPORT OF THE THEORY OF ROTATION OF THE WESTERN PART OF THE ISLAND OF NEWFOUNDLAND

By R. F. BLACK

Geological Survey of Canada, Ottawa

IN 1959 two papers were published regarding the possible post-Carboniferous rotation of the Island of Newfoundland relative to the mainland of North America. Both papers were based on palæomagnetic data from rocks collected in Newfoundland and on the mainland of North America. Nairn *et al.*[1] suggested an anticlockwise rotation of Newfoundland relative to the mainland on the basis of his palæomagnetic measurements of the Mississippian strata of western Newfoundland as compared with similar data reported by others[2] on the palæomagnetism of Mississippian and Pennsylvanian strata of western and central United States. The virtual geomagnetic poles[3] derived from each of these two groups of data were separated by approximately 20 deg. on a great circle at the Earth's surface. A 20-deg. clockwise rotation of Newfoundland was sufficient to eliminate this difference. In the light of Nairn's[1] data and of his own palæomagnetic measurements on Mississippian and Pennsylvanian strata from the mainland of eastern Canada, Du Bois[4] concluded that their was no evidence for relative rotation since Carboniferous time between Newfoundland and the mainland of eastern Canada. His conclusion was based on the fact that the declinations of the mean magnetization vectors for the two groups did not differ significantly relative to a common co-ordinate system.

In 1962 I made a collection of oriented rock samples from Palæozoic sedimentary strata of western Newfoundland (locations shown in Fig. 1). The collection included samples from the Mississippian Codroy Group from along the south-east shore of St. Georges Bay[5], the Clam Bank Group of Lower Devonian age from the north-west shore of the Port au Port Peninsula[6], and the Bradore Formation of Lower Cambrian age from the Highlands of St. John[6]. From each of the samples, two oriented cubical specimens were cut and the directions of magnetization measured with an astatic magnetometer put at my disposal by the Dominion Observatory of Canada. The directions of magnetization were not significantly affected when treated in decreasing-amplitude alternating magnetic fields up to 300 œrsteds (peak value). A mean direction of magnetization for each sample was computed from two specimens. A mean formation or group direction of magnetization was determined to calculate the virtual geomagnetic pole position. Table 1 includes these results and their statistics[7] as well as the mean directions of magnetization, pole positions, and statistics for groups of rocks collected by officers of the Geological Survey of Canada during 1955, 1957, 1958 and 1960 from the mainland of south-eastern Canada (Fig. 1). The latter samples have been treated as above and were found to be practically unaffected by the alternating magnetic field.

The results of the Bathurst[8], Bonaventure[8], and Kennebecasis[9] Formations given in Table 1 were combined by Du Bois[4] to give a mean direction of magnetization for the mainland Carboniferous rocks. The results determined by the Geological Survey of Canada for the red beds of Prince Edward Island (PEI)[10], the pre-Pictou sandstones[11], the volcanic rocks and sediments of the Perry Formation[12], and the sandstones of the Ratcliffe Brook Formation[9], are reported here for the first time.

Comparison of the virtual geomagnetic pole positions from Newfoundland with those from the mainland (Fig. 2) shows that Newfoundland has not rotated relative to the mainland of eastern Canada since Carboniferous time, in

Table 1. SUMMARY OF PALÆOMAGNETIC DATA FOR SOME PALÆOZOIC ROCKS OF EASTERN CANADA

Formation	Geological age	No. on Figs. 1 and 2	Magnetic direction					Pole position			
			D	*I*	*N*	α95	*K*	Lat. N	Long. E	dψ	dχ
Red Beds, P.E.I.‡	Permo Carboniferous	15	174·7	+8·1	—	7·0	—	39·0	123·2	3·7	7·4
Red Beds, Eastern P.E.I.	,, ,,	14	176·8	+5·5	59	5·5	12·5	40·9	121·7	2·7	5·5
Red Beds, Central P.E.I.	,, ,,	13	170·9	+7·2	69	5·5	10·6	39·5	128·3	2·8	5·5
Red Beds, Northwest P.E.I.	,, ,,	12	170·0	+5·3	26	6·1	22·6	39·8	129·0	3·1	6·1
Bathurst†	Pennsylvanian	11	162·3	+15·6	10	9·3	21·8	31·9	134·8	4·9	9·5
Bonaventure†	Penn. and/or Miss.	10	166·1	+13·1	22	6·8	21·7	34·0	130·8	3·5	6·9
Pre-Pictou Sandst.†	,, ,, ,,	9	161·5	+30·5	8	15·4	13·8	23·5	133·4	9·6	17·2
Codroy Group	Mississippian	8	174·5	+23·4	32	10·1	7·3	29·6	127·2	5·7	10·7
,, ,,*	,,	7	166·0	+8·0	9	8·4	38·0	43·0	139·0	4·3	8·5
Kennebecasis†	,,	6	161·3	+33·3	14	9·6	18·1	21·7	133·2	6·2	10·9
Perry Vol.	Upper Devonian	5	183·5	+33·8	16	15·1	7·0	26·4	109·3	9·8	17·2
Perry Sed.	,, ,,	4	173·7	+19·5	36	8·5	8·9	34·7	120·5	4·6	8·8
Clam Bank Group	Lower Devonian	3	337·5	−19·8	18	11·4	10·2	28·0	146·2	6·2	11·9
Ratcliffe Brook	Lower Cambrian	2	168·1	+53·2	12	14·9	9·5	10·0	124·0	14·3	20·6
Bradore	,, ,,	1	151·2	+44·5	10	9·4	27·4	9·0	148·8	7·4	11·8

D, mean declination of magnetization; *I*, mean inclination of magnetization; *N*, number of samples used for each formation; α95, radius of the circle of confidence at the 95 per cent level of probability; *K*, estimate of precision parameter; dψ, dχ, semi-axes of the oval of confidence. * As reported by Nairn *et al.* (ref. 1); † *D* and *I*, computed for co-ordinates 66·0 W., 48·0 N.; ‡ as reported by Roy (ref. 14), 60 specimens from 17 sites.

©1964 Nature Publishing Group

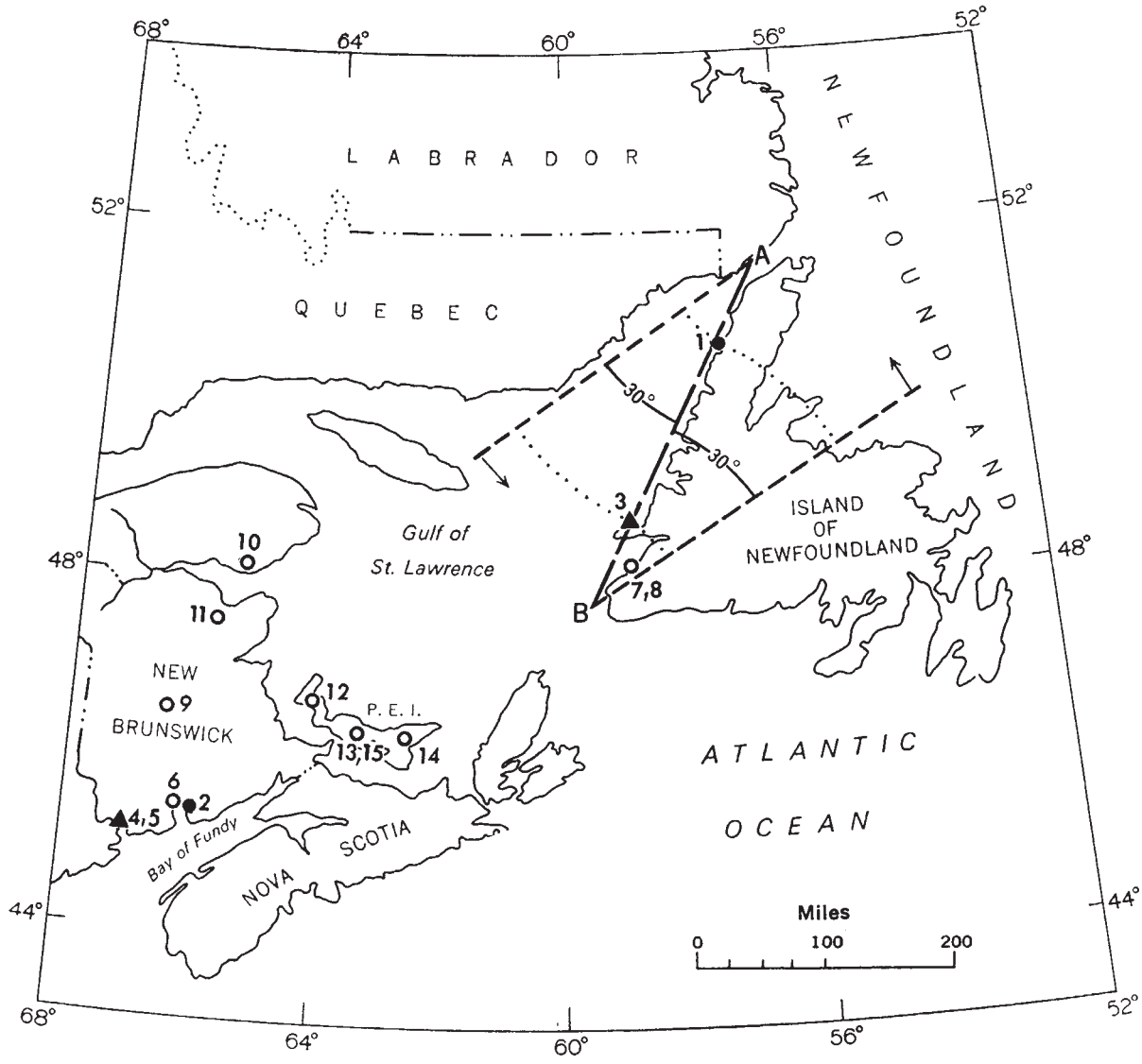


Fig. 1. Index map showing collecting localities of Cambrian, Devonian and Carboniferous rocks of eastern Canada. Lambert conformal conic projection

agreement with Du Bois's conclusion. There is, on one hand, a near coincidence of the Carboniferous poles for Newfoundland with respect to those for the mainland, but on the other hand significant differences do exist between the virtual pole positions for the Devonian and Cambrian rocks from Newfoundland and those from the mainland of eastern Canada of the same geological period. The virtual pole positions for the Lower Devonian Clam Bank Group and the Lower Cambrian Bradore Formation of western Newfoundland plot, approximately 30 deg. east of, and at the same approximate latitudes respectively as, those determined for the Upper Devonian Perry Formation and the Lower Cambrian Ratcliffe Brook Formation from southern New Brunswick.

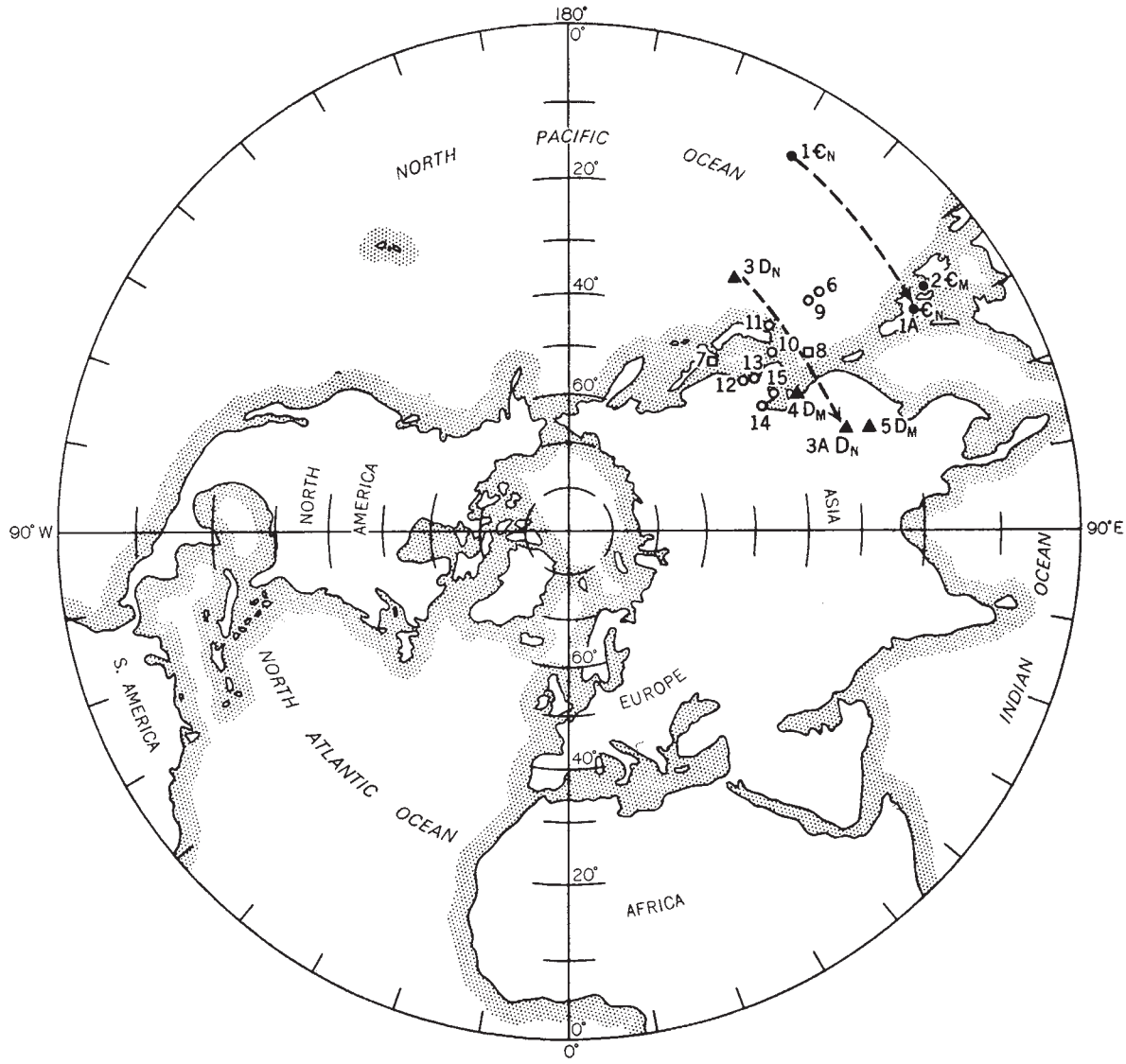
If Newfoundland is rotated 30 deg. horizontally in a clockwise direction, the Devonian and Cambrian poles for both Newfoundland and the mainland become nearly coincident. Such a rotation could bring the western shoreline of Newfoundland close to and parallel with the northern coastline of the Gulf of St. Lawrence, as this is the approximate angle that exists to-day between the two coastlines (Fig. 1). The poles for the Newfoundland Devonian and Cambrian rocks when recalculated on the basis of the addition of 30 deg. to the declinations of the

©1964 Nature Publishing Group

mean magnetization vectors move close to their mainland counterparts (Fig. 2).

The position of Newfoundland relative to the mainland from Cambrian to Lower Devonian time cannot be determined solely on the basis of the foregoing palæomagnetic data because a unique pivot point cannot be ascertained. If, for purposes of argument, it is assumed that Newfoundland pivoted about a point near the Strait of Belle Isle (position *A*, Fig. 1) then the western shore of Newfoundland must have moved away from the mainland during Middle and/or Upper Devonian time. Similarly an anticlockwise rotation about the southern tip of western Newfoundland (position *B*, Fig. 1) during this time implies movement of the western shore of Newfoundland in a north-westward direction to its present position. Both possibilities involve an anticlockwise rotation and suggest two distinct parallel positions (schematically shown in Fig. 1) for the western shoreline of Newfoundland during Cambrian to Lower Devonian time; the one with the present shorelines close together, the other with open sea between. The geographical co-ordinates of the collection localities in Cambrian to Lower Devonian times, therefore, depend on the former position of the western coastline of Newfoundland.

The relative displacement of one land-mass with respect to another, by a hypothetical rotation, may be determined by observing the relative position of the virtual pole positions obtained as a result of rotation about selected pivot points when compared with those from the fixed land-mass. This may be used as a criterion for obtaining the point of rotation only if the translation involved will alter the co-ordinates of the collection locality by an amount such that the ovals of confidence about the virtual pole positions do not intersect each other. The pivot point can be determined by observing which of the pole positions, as a result of rotation, is nearer that of the fixed formation of the same age. Two possible sets of co-ordinates for the Cambrian and Devonian collection



Cambrian pole positions for Newfoundland and mainland Canada . . . . . $€_N$ $€_M$ ●

Devonian pole positions for Newfoundland and mainland Canada . . . . . D_N D_M ▲

Carboniferous pole positions for Newfoundland and mainland Canada . . . . . □, ○

Numbers refer to formations listed in Tables 1 and 2

Fig. 2. Polar stereographic projection of northern hemisphere showing Palæozoic pole positions as determined from rocks of eastern Canada and the effect of a 30-deg. anticlockwise rotation of Newfoundland during Devonian time

©1964 Nature Publishing Group

Table 2. SUMMARY OF THE CORRECTED MEAN DIRECTIONS OF MAGNETIZATION AND THE CO-ORDINATES OF THE COLLECTION LOCALITIES AND POLE POSITIONS FOR DEVONIAN AND CAMBRIAN ROCKS ASSUMING NEWFOUNDLAND UNDERWENT A 30-DEG. ROTATION ABOUT POINTS *A* OR *B* ON FIG. 1

Formation	Geological age	Rotation about	Collection locality Lat. N	Collection locality Long. W	Corrected mean direction *D*	Corrected mean direction *I*	Pole position Lat. N	Pole position Long. E
Clam Bank	Lower Devonian	*A*	49·8	60·8	6·0	−17·5	30·0	112·5
Bradore	Lower Cambrian	*A*	51·2	57·8	180·5	+44·0	13·0	122·0
Clam Bank	Lower Devonian	*B*	48·3	58·3	8·5	−20·5	30·5	111·0
Bradore	Lower Cambrian	*B*	49·5	55·4	184·0	+47·0	12·0	120·0
Clam Bank	Lower Devonian	Mean of rotations *A* and *B*, number 3*A* on Fig. 2					30·5	112·0
Bradore	Lower Cambrian	Mean of rotations *A* and *B*, number 1*A* on Fig. 2					12·5	121·0

localities are given in Table 2 along with the corresponding virtual pole positions. It is impossible, however, to choose between *A* and *B* as the rotation point because the ovals of confidence about the pole positions are too large compared with the distances between the collection localities. A mean pole position for each of the Cambrian and Devonian Formations of Newfoundland has therefore been calculated from the pole positions determined by rotation about *A* and *B*. These are plotted in Fig. 2.

The geology and tectonics of the Appalachian Region of Canada[13] can be interpreted to favour anticlockwise rotation of Newfoundland during later Devonian time. Two broad belts of volcanic-bearing strata, one of Cambro-Ordovician age and the other of Siluro-Devonian age, each flanked by sediments of similar age occur on the mainland and on the Island. They would appear to have been displaced relative to each other, and a clockwise rotation of Newfoundland would reduce the divergence in their structural trends. On the other hand, the belt of deformed and relatively thick Carboniferous strata extends in a nearly straight line through the mainland and western Newfoundland. The proposed rotation must therefore have occurred in Middle to Late Devonian time as suggested by the palæomagnetic data.

The palæomagnetic data presented here are useful in suggesting the direction and approximate angle of relative rotation as well as the time of its occurrence. They suggest that, from Lower Cambrian to Lower Devonian time, Newfoundland remained fixed with respect to mainland Canada and was not rotated during the Ordovician Taconic orogeny. The 30-deg. anticlockwise rotation of at least the western part of Newfoundland where the samples were collected may therefore be identified with the Acadian orogeny of Middle to Late Devonian time. Moreover, the Late Palæozoic orogeny in the Canadian Appalachian Region[13] did not contribute significantly to the rotation.

I thank Mr. G. N. Freda for his assistance.

1 Nairn, A. E. M., Frost, D. V., and Light, B. G., *Nature*, **183**, 596 (1959).
2 Creer, K. M., Irving, E., and Runcorn, S. K., *Proc. Roy. Soc.*, A, **250**, 144 (1957).
3 Cox, A., and Doell, R. R., *Bull. Geol. Soc. Amer.*, **71**, 645 (1960).
4 Du Bois, P. M., *Nature*, **184** (B.A.), 63 (1959).
5 Bell, W. A., *Geol. Surv. Canada. Bull.*, 10 (1948).
6 Schuchert, C., and Dunbar, C. O., *Geol. Soc. Amer., Mem.* 1 (1934).
7 Fisher, R. A., *Proc. Roy. Soc.*, A, **217**, 295 (1953).
8 Alcock, F. J., *Geol. Surv. Canada, Mem.* 183 (1935).
9 Alcock, F. J., *Geol. Surv. Canada, Mem.* 216 (1938).
10 Milligan, G. C., *Dept. Indust. and Natural Resources*, Province of Prince Edward Island (1949).
11 Anderson, F. D., and Poole, W. H., *Geol. Surv. Canada, Map* 37 (1959).
12 Alcock, F. J., *Geol. Surv. Canada, Preliminary Map* 46—2.
13 Neale, E. R. W., Beland, J., Potter, R. R., and Poole, W. H., *Canad. Inst. Mining Metall. Bull.*, **54**, 687 (1961).
14 Roy, J. L., *Geophys. J. Roy. Astron. Soc.*, **8**, No. 2 (1963).

ANALOGUE COMPUTATION SIMULATED DIGITALLY

By J. L. DINELEY and DR. L. MOLYNEUX

Departments of Electrical Engineering and Physics, University of Newcastle upon Tyne

THERE is a bias towards solving engineering problems, particularly these involving non-linear differential equations with an analogue rather than digital computer. This is due more to the way in which the problem is set up than to any inherent advantage in the method of calculation. In the analogue machine the problem is arranged as a model where the physical characteristics of the component parts of the real system are represented in the form of electronic computing elements. There is a close correspondence between the model and its physical counterpart which makes it easy to translate measurements and ideas from one to the other. But it has a number of practical limitations, the most important of which are as follows:

(*a*) In general the accuracy of computation is low, partly because of the difficulty of setting and maintaining values of the passive components and partly because of the inadequacies of the computing units, for example, operational amplifiers, which, in addition to having theoretical limitations such as finite gain, may also introduce spurious signals into the system. Again, certain operations, for example multiplication and division, are difficult to carry out with high accuracy.

Non-linear relationships are difficult to synthesize and often introduce limitations such as speed of calculation, as well as adding spurious signals.

(*b*) Problems take a considerable time to set up, and during the setting-up time the computer cannot be used for other work.

(*c*) Since each computing unit introduces errors it is advisable to devise a model which uses the minimum of computing equipment. This takes time and may also detract from the relationship between the analogue and the real system.

(*d*) As the computing units are linear over only a limited voltage range the system must be scaled so that this range is nowhere exceeded during the computation.

A digital computer has none of these limitations, and the challenge is therefore to devise a programming method which will allow the conceptual advantages of the analogue computer to be realized with a digital machine.

The elaborate digital machines now coming into use can perform calculations at great speed and have ample storage capacity. The sophistication and high speed have achieved two important results.

(1) The programme can be written in a form (*ALGOL*, for example) that is convenient and easy to write without markedly slowing the speed of calculation.

(2) Even with programmes written in such a convenient form the cost per computation (multiplication, addition, etc.) is low.

©1964 Nature Publishing Group